WINC1500 Software

Release Notes

VERSION : **19.7.10**

DATE : **JUN, 2025**

Abstract

This document presents an overview of the WINC15x0 firmware release version 19.7.10, and corresponding driver.

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1 Introduction

This document describes the WINC15x0 version 19.7.10 release package. This is a refresh release that includes multiple fixes and enhancements since the 19.7.7 release.

The release package contains all the necessary components (binaries and tools) required to make use of the latest features including tools, driver code and firmware binaries.

1.1 Firmware readiness

Microchip Technology Inc. considers version 19.7.10 firmware to be suitable for production release.

2 Release summary

2.1 Auditing information

Master Development Ticket	:	https://jira.microchip.com/projects/W1500/versions/81715
Release Repository	:	Wifi_M2M
Source Branch	:	/branches/rel_1500_19.7.10
Subversion Revision	:	r20756

2.2 Version information

WINC Firmware version	:	19.7.10
Host Driver version	:	19.7.10
Minimum driver version	:	19.3.0

Please note that the SVN revision advertised in the firmware serial trace will be **20755**.

```
(10)NMI M2M SW  VER 19.7.10 REV 20755
(10)NMI MIN DRV VER 19.3.0
(10)FW URL branches/rel_1500_19.7.10
(10)Built Jun 27 2025   16:14:50
```

2.3 Released components

The release contains documentation, sources and binaries.

2.3.1 Documentation overview

The Application manuals, Release notes and Software API guides can be found in the `doc/` folder of the release package.

Release Notes:

This document

Software APIs:

WINC1500_IoT_SW_APIs.chm

2.3.2 Binaries and programming scripts

The main WINC15x0 firmware binary is located in the `firmware` directory and named `m2m_aio_3a0.bin`. This can be flashed to a WINC device using, for example, a serial bridge application available from ASF.

An OTA image is provided in the `ota_firmware` directory named `m2m_ota_3a0.bin`.

2.3.3 Sources

Source code for the host driver can be found under the `src/host_drv` directory.

Source code for the tools, including crypto_lib, can be found under the `src/Tools` directory.

2.4 Release Comparison

Features in 19.7.7	Changes in 19.7.10
Wi-Fi STA	
• IEEE 802.11 b/g/n. • OPEN security. • WPA Personal Security (WPA1/WPA2). • WPA Enterprise Security (WPA1/WPA2) supporting: EAP-TTLSv0/MS-Chapv2.0 EAP-PEAPv0/MS-Chapv2.0 EAP-PEAPv1/MS-Chapv2.0 EAP-TLS EAP-PEAPv0/TLS EAP-PEAPv1/TLS • 'Fragattack' countermeasures	• Allow enabling/disabling of specific Phase 1 WPA Enterprise methods. • Added EAPOLv3 support to WPA Enterprise Security. • Fixed code that saves connection info to WINC flash upon successful connection to ensure it doesn't perform unnecessary flash writes
Wi-Fi Hotspot	
• Only ONE associated station is supported. After a connection is established with a station, further connections are rejected. • OPEN and WPA/WPA2 security modes. • The device cannot work as a station in this mode (STA/AP Concurrency is not supported). • 'Fragattack' countermeasures.	No change
WPS	
The WINC1500 supports the WPS protocol v2.0 for PBC (Push button configuration) and PIN methods.	No change
TCP/IP Stack	
The WINC1500 has a TCP/IP Stack running in firmware side. It supports TCP and UDP full socket operations (client/server). The maximum number of supported sockets is currently configured to 11 divided as: • 7 TCP sockets (client or server). • 4 UDP sockets (client or server).	No change
TLS	
• Support TLS v1.2. • Client and server modes. • Mutual authentication in client mode.	• The "critical" field of x.509 certificate extensions is now correctly handled

Features in 19.7.7	Changes in 19.7.10
• Subject Alternate Name support in client mode • X509 certificate revocation scheme. • SHA384 and SHA512 support in X509 certificates processing. • Integration with ATECC508 (ECDSA and ECDHE support). • Supported cipher suites are: TLS_RSA_WITH_AES_128_CBC_SHA TLS_RSA_WITH_AES_128_CBC_SHA256 TLS_RSA_WITH_AES_256_CBC_SHA TLS_RSA_WITH_AES_256_CBC_SHA256 TLS_DHE_RSA_WITH_AES_128_CBC_SHA TLS_DHE_RSA_WITH_AES_128_CBC_SHA256 TLS_DHE_RSA_WITH_AES_256_CBC_SHA TLS_DHE_RSA_WITH_AES_256_CBC_SHA256 TLS_RSA_WITH_AES_128_GCM_SHA256 TLS_DHE_RSA_WITH_AES_128_GCM_SHA256 TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 (requires ECC508) TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256 (requires ATECC508)	• Ensure Basic Constraint is checked in server certificate chain
Networking Protocols	
DHCPv4 (client/server) DNS Resolver IGMPv1, v2. SNTP	No change
Power saving Modes	
• M2M_PS_MANUAL • M2M_PS_DEEP_AUTOMATIC	No change
Device Over-The-Air (OTA) upgrade	
• Built-in OTA upgrade available. • Backwards compatible as far as 19.4.4, with the exception of: - Wi-Fi Direct (removed in 19.5.3) - Monitor mode (removed in 19.5.2)	No change
Wi-Fi credentials provisioning via built-in HTTP server	
Built-in HTTP/HTTPS (TLS server mode) provisioning using AP mode (Open, or WPA/WPA2 secured).	No change

Features in 19.7.7	Changes in 19.7.10
Ethernet Mode (TCP/IP Bypass)	
Allow WINC1500 to operate in WLAN MAC only mode and let the host send/receive Ethernet frames.	No change
ATE Test Mode	
Embedded ATE test mode for production line testing driven from the host MCU.	No change
Miscellaneous features	
	No change

3 Test Information

Please refer to ticket W1500-960 for full details.

Testing was performed against the release candidate 19.7.10 against the following configuration(s):

H/W Version : WINC1510 Xplained module

Host MCU : ATSAMD21-Xplained

The following testing was performed in both open air and shielded environments;

1. General functionality including:
    1. HTTP Provisioning
    2. Station Mode
    3. AP Mode
    4. IP (TCP and UDP client and server)
    5. HTTP POST/GET
    6. WPS (PIN and PushButton methods)
    7. Over-The-Air (OTA) update functionality and robustness (with and without TLS)
2. TLS functionality including:
    1. RSA cipher-suites:
        i. TLS_RSA_WITH_AES_128_CBC_SHA
        ii. TLS_RSA_WITH_AES_128_CBC_SHA256
        iii. TLS_RSA_WITH_AES_128_GCM_SHA256
        iv. TLS_DHE_RSA_WITH_AES_128_CBC_SHA
        v. TLS_DHE_RSA_WITH_AES_128_CBC_SHA256
        vi. TLS_DHE_RSA_WITH_AES_128_GCM_SHA256

        Testing uses 1024-bit, 2048-bit and 4096-bit server certificates, with a chain of 7 certificates of varying key lengths (1024,2048 and 4096 bit) leading to a 2048-bit root certificate.
    2. ECDSA ciphersuites:
        i. TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256
        ii. TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256

        Testing uses a NIST standard ECC P256 prime curve server certificate with two chains, one leading back to an ECDSA root certificate and the other leading to an RSA root certificate.
    3. Client authentication
3. Performance under interference
4. TCP/IP stack robustness testing
    1. Using an internal implementation of IPerf.
    2. Verification of multi socket functionality

4 Known issues

ID	Description
W1500-63	Occasionally WINC15x0 fails to receive an individual UDP broadcast frame when in M2M_PS_DEEP_AUTOMATIC powersave mode. Recommended workaround: **Use M2M_NO_PS powersave mode if reliability is preferred for UDP broadcast frames. Otherwise ensure the overlying protocol can handle the odd missing frame.**
W1500-108	The WINC15x0 cannot handle two simultaneous TLS handshakes, due to memory constraints. Recommended workaround: **When attempting to open two secure sockets in STA mode, the application should wait to be notified of the first one completing (succeeding or failing) before attempting the second one.**
W1500-325	1% of Enterprise conversations fail due to the WINC15x0 not sending an EAP response. The response is prepared and ready to send but does not appear on the air. After 10 seconds the firmware times-out the connection attempt and the application is notified of the failure to connect. Recommended workaround: Configure the authentication server to retry EAP requests (with interval < 10 seconds). **The application should retry the connection request when it is notified of the failure.**
W1500-369	When connected to certain access points, the WINC15x0 sometimes fails to roam when the access point changes channel. The issue is seen with these access points: Linksys E2500, Linksys E4200, Linksys 6500. The failures to roam are due to two issues: 1. Sometimes the access point takes a long time to start sending beacons or probe responses on the new channel, so it is not discoverable. 2. Sometimes the access point does not initiate the 4-way handshake (for WPA/WPA2 PSK reconnections). Recommended workaround: **On reception of M2M_WIFI_DISCONNECTED event, the application should attempt to discover the access point using `m2m_wifi_request_scan()` API.**

W1500-387	If an AP uses an 802.11 ACK policy of “No Ack”, then the WINC15x0 sometimes fails to receive 802.11b frames. Recommended workaround: **Avoid using an ACK policy of “No Ack”. If “No Ack” is used, ensure frames are sent at 802.11g or higher rates.**
W1500-397	70% of Enterprise connection requests fail with a TP Link Archer D2 access point (TPLink-AC750-D2). The access point does not forward the initial EAP Identity Response to the authentication server. The issue is bypassed by PMKSA caching (WPA2 only), so reconnection attempts will succeed. Recommended workaround: **The application should retry the connection request when it is notified of the failure.**
W1500-402	Occasionally during AP provisioning, after entering the credentials of the AP to connect to and pressing “connect”, an error will be returned even though provisioning was successful and the connection proceeds. Recommended workaround: **Add a delay in the application between receiving the provisioning info and connecting to the AP. Ignore the “Request Failed” message.**
W1500-699	When using a driver pre – 19.6.0 with this firmware, upon failure to obtain a DHCP address the WINC will not trigger a WiFi Disconnection and notify the driver of the failure. Recommended workaround: **In this case of an older driver running with later firmware, the application should monitor the time taken to obtain a DHCP address, if it takes too long then it can decide whether to disconnect and try again.**
W1500-854	When sustaining a maximum throughput TLS RX stream using a TLS record size of 16K, if another maximum throughput TLS RX stream with the same record size is established, the streams sometimes become stuck and fail to transfer any more data. Recommended workaround: **Try to avoid using 2 concurrent TLS streams receiving 16K records. If this scenario is used and the streams become stuck, close and reopen the sockets and try again.**
W1500-836	TLS Server mode does not work with ECDHE ciphersuites. Recommended workaround: **Disable ECDHE ciphersuites before starting TLS Server.**
W1500-865	When running a sustained high throughput TCP RX stream over a long period (>1 hour), the WINC can sometimes miss enough beacons from the AP to trigger link loss detection and will disconnect from the AP. Recommended workaround: **If the WiFi connection is closed by the WINC during a TCP RX stream, reconnect and re-open the socket and try again.**

5 Fixes and enhancements

These are the major fixes and enhancements since the previous released version (19.7.7).

5.1 Issues fixed

ID	Description
W1500-870	**During second phase of Enterprise authentication, long EAP Responses undergo double-fragmentation** For PEAPv0, double-fragmentation (i.e. fragmenting at both the EAP-TLS and PEAP layers) causes inter-operability issues, in particular it prevents authentication with FreeRadius servers. Fixed: Fragmentation thresholds adjusted to ensure that EAP Responses are not fragmented at the PEAP layer
W1500-873	**NULL BSSID set in frames that follow a deauthentication** After a deauthentication frame has been sent to the WINC1500, there is a short window where the WINC1500 may attempt to send subsequent frames with a NULL destination address until it has processed the deauthentication and send out its own deauthentication frame in response. Fixed: Modifications made to the firmware to close the window in which this can happen.
W1500-879	**In SoftAP mode, DHCP request messages could sometimes be incorrectly filtered out.** This was due to an incorrect check when determining if a received message was unicast or multicast, occasionally causing decryption to fail because the wrong key was used. Fixed: Corrected the logic to ensure the unicast/multicast check is always accurate
W1500-892	**New WiFi connections disallowed after a failed Default Connect** When a Default Connection fails (m2m_wifi_default_connect()), in some instances the WINC1500 will refuse all new connection attempts. Fixed: Ensure new WiFi connections can be attempted after a default connect failure
W1500-929	**TLS stack does not check CA Basic Constraint in the server certificate chain** As per RFC 5280 section 4.2.19, a TLS client should check the CA Basic Constraint in certificates in the received server certificate chain. This field should be set in all certificates except the end entity server certificate. The WINC1500 was not checking this field in the certificate chain. Fixed: The WINC1500 now checks this field in the certificate chain
W1500-944	**A small number of devices may exhibit occasional initialization errors** An adjustment to the clock switching sequence during RF initialization ensures these parts remain stable during startup.

W1500-946	**Connection parameter info in flash is updated unnecessarily** The Connection Parameter information should only be updated in flash if the connection parameters have changed. However, WINC1500 was updating parts of the Connection Parameter info when there had been no changes, resulting in unnecessary flash writes. Fixed: Code refactored to only write to flash when necessary (e.g. connection parameters (such as SSID, passphrase) are changed)
W1500-948	**Inappropriate casting in m2m_ssl_retrieve_cert()** The casting of variable pu16Curve was leading to corruption of the upper two bytes. Fixed: Code adjusted to resolve casting issue
W1500-949	**Correctly parse and handle the Critical Field of x.509 certificate extensions** The x.509 v3 SAN extension (see rfc5280) includes a boolean designating whether the extension in a certificate is designated as either critical or non-critical. WINC1500 does not parse this extension correctly in this or other x.509 extensions. Fixed: Code adjusted to ensure X.509 extensions are correctly processed
W1500-953	**WPA Enterprise issue where EAPOL exchanges with certain APs result in connection failure** It has been observed that some APs send an EAP Identity Request message before receiving an EAP Start from the WINC1500, then send another Identity Request after the WINC1500 sends its EAP Start. This sequence can cause the EAP connection to become out of sync and fail. Fixed: Only send an EAP Start message after waiting to first see if the AP automatically sends an EAP Identity request straight after association.
W1500-954	**If a single RX buffer contains more than 7 TLS records, the excess records are dropped and do not reach the host** This restriction limits the usability of protocols such as MQTT that may require the device to receive many short TLS records in quick succession. Fixed: Dynamically allocate resources for managing data reception. Now up to 51 TLS records may be received in quick succession. In order to ensure smooth operation, the host application should read the data regularly to avoid the number of queued records from approaching or exceeding this limit of 51.
W1500-962	**WINC1500 can run out of memory when handling very long TLS certificate chains** TLS certificate chains containing several certificates can result in the WINC1500 running out of memory as it tries to validate the chain Fixed: Improve memory allocation for TLS certificates

5.2 Enhancements

W1500-814	**Add support for EAPOLv3 messages in the WPA Enterprise handshake** WINC1500 would previously reject EAPOLv3 messages from an AP, resulting in failure to connect to the WPA Enterprise network

W1500-871	**Specific Enterprise Phase 1 method configuration**
	It is now possible to specify particular Phase 1 methods for WPA Enterprise via the `m2m_wifi_1x_set_option()` API.
	
	For example, the following code sets the Phase 1 method to only EAP-TLS, no other method is allowed:
	
	`tenu1xPhase1Method option_value;`
	`option_value = WIFI_1X_PHASE1_METHOD_EAP_TLS;`
	`m2m_wifi_1x_set_option(WIFI_1X_PHASE1_METHOD, &option_value, sizeof(option_value));`
	
	It is possible to get the currently configured option using the function `m2m_wifi_1x_get_option()`.
	
	Please refer to the API documentation for more information.

W1500-958	**Improved consistency in the generation of cryptographic Wi-Fi elements** Previously, when the WINC1500's first use after startup was Wi-Fi Hotspot mode, cryptographic Wi-Fi elements for that mode were generated using a sub-optimally configured Random Number Generator. This has been updated so that the RNG is now properly configured when used for cryptographic Wi-Fi elements in all scenarios.

6 Appendix A – TLS Root certificates

The WINC1500 19.7.10 module comes with a preselected selection of TLS root certificates that will allow a TLS connection to be established with a range of internet TLS servers out of the box.

Note that in 19.7.10 the Baltimore Cyber Trust Root certificate has expired and been replaced with Digicert Global Root G2.

These preselected certificates are described in 6.1

6.1 TLS root certificates

Issuer	Filename	Expiry	Public Key	Signature Alg.	Notes
Amazon Root CA 1	AmazonRootCA1.cer	17 January 2038 01:00:00	RSA (2048 bits)	SHA256RSA	AWS Cloud
DigiCert Global Root G2	DigiCertGlobal-RootG2.cer	15 January 2038 13:00:00	RSA (2048 bits)	SHA256RSA	Azure Cloud
DigiCert High Assurance EV Root CA	DigiCert.cer	10 November 2031 01:00:00	RSA (2048 bits)	SHA1RSA	
DigiCert High Assurance EV Root CA	DigiCertSHA2.cer	22 October 2028 13:00:00	RSA (2048 bits)	SHA256RSA	
Entrust Root Certification Authority	EnTrust.cer	27 November 2026 21:53:42	RSA (2048 bits)	SHA1RSA	
GlobalSign Root CA	GlobalSignRoot.cer	28 January 2028 13:00:00	RSA (2048 bits)	SHA1RSA	
Internet Security Research Group Root X1	isrgrootx1.cer	04 June 2035 12:04:38	RSA (4096 bits)	SHA256RSA	LetsEncrypt
QuoVadis Root CA 2	QuoVadis_Root.cer	24 November 2031 19:23:33	RSA (4096 bits)	SHA1RSA	
VeriSign Class 3 Primary Certification Authority	VeriSign.cer	17 July 2036 00:59:59	RSA (2048 bits)	SHA1RSA	

7 Terms and Definitions

Term	Definition
AES	Advanced Encryption Standard
AJAX	Asynchronous JavaScript and XML
AKM	Authentication and Key Management
ARP	Address Resolution Protocol
ATE	Automated Test Equipment
BSS	Basic Service Set
CBC	Cyclic Block Chaining
DHCP	Dynamic Host Control Protocol
DHE	Diffie-Hellman Ephemeral
DNS	Domain Name Server
DTIM	Directed Traffic Indication Map
EAP	Extensible Authentication Protocol
EAPOL	EAP Over LAN
ECC	Elliptic Curve Cryptography
ECDHE	Elliptic Curve Diffie-Hellman Ephemeral
ECDSA	Elliptic Curve Digital Signature Algorithm
EEPROM	Electrically Erasable Programmable Read Only Memory
ESD	Electrostatic Discharge
EVM	Error Vector Magnitude
HIF	Host Interface
HTTP	Hypertext Transfer Protocol
IEEE	Institute of Electronic and Electrical Engineers
MAC	Media Access Control
OTA	Over The Air update
PEAP	Protected Extensible Authentication Protocol
PLL	Phase Locked Loop
PMK	Pair-wise Master Key
PSK	Pre-shared Key
QAM	Quadrature Amplitude Modulation
RSA	Rivest-Shamir-Adleman (public key cryptosystem)
RSN	Robust Security Network
RSSI	Receive Strength Signal Indicator
SHA	Secure Hash Algorithm
SNTP	Simple Network Time Protocol
SPI	Serial Peripheral Interface
SSID	Service Set Identifier
SSL	Secure Sockets Layer
TCP	Transmission Control Protocol
TIM	Traffic Indication Map
TLS	Transport Layer Security

Term	Definition
WEP	Wired Equivalent Privacy
WINC	Wireless Network Controller
WLAN	Wireless Local Area Network
WMM™	Wi-Fi Multimedia
WMM-PS™	Wi-Fi Multimedia Power Save
WPA™	Wi-Fi Protected Access
WPA2™	Wi-Fi Protected Access 2 (same as IEEE 802.11i)